

SPECTRA: Stellar Parameter Estimation and Calculation Tools for Research and Analysis

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Summary

Determining fundamental stellar parameters — such as mass, age, luminosity, effective temperature (T_{eff}), radius, and distance — is central to observational and theoretical astrophysics (Serenelli et al., 2021). **SPECTRA**¹ (*Stellar Parameter Estimation and Calculation Tools for Research and Analysis*, v1.0.0_build_160726) is an open-source Python tool designed to optimize and automate stellar mass determination and diagnostic modeling for stars in clusters, with a particular focus on young, low-mass stars.

The software integrates data preprocessing, missing data imputation via k -Nearest Neighbors (k -NN) and iterative techniques, theoretical isochrone fitting (IsocFit), machine-learning regression modeling (Mass-Magnitude Modeling), and interactive visualization tools (Baraffe et al., 2015; Serenelli et al., 2021).

Statement of need

In Astrophysics, estimating the mass of single stars outside binary systems relies heavily on theoretical stellar evolution models and mass-luminosity or mass-magnitude relations (MLR/MMR) (Benedict et al., 2016; Serenelli et al., 2021). Taking advantage of age and metallicity homogeneity of stars belonging to clusters, isochrone or stellar track fitting on color magnitude diagrams proves to be a useful tool to obtain simultaneously stellar ages and mass. However, it depends on a well defined main sequence which in turn depends on high quality data and on the age of the cluster. Younger clusters have a great number of pre main sequence (PMS) stars that lie above the main sequence making the fitting more difficult. Moreover, PMS stars exhibit high observational uncertainties due to differential interstellar extinction, age dispersion, and incomplete data vectors (Herczeg & Hillenbrand, 2015). Data quality and constraints, differences between stellar models affect the final results. Stellar structure and evolutionary models differ in the adopted input physics and this impact the isochrone fitting results (Lebreton et al., 2014). On the other hand, MLR or MMR are model independent and rely on gathering the best available photometric and spectroscopic data for systems that can provide direct mass values as, for example, double-lined eclipsing binary systems. Using MLR/MMR requires goodness-of-fit methods that can be statistical in nature or that can rely on the use of machine learning techniques. SPECTRA uses both isochrone fitting and MMR relations to calculate mass values of stars belonging to stellar clusters.

SPECTRA addresses these challenges by offering a unified, user-friendly desktop application built

¹The development of this software and its underlying research framework were initiated and primarily executed at the Universidade Estadual de Santa Cruz (UESC), with final updates and infrastructure migration supported by the Universidade Estadual Paulista (UNESP).

with `ttkbootstrap`. It enables researchers and educators to: 1. Automatically preprocess, filter, and impute incomplete photometric datasets. 2. Execute automated local interpolation over theoretical evolutionary grids (e.g., Siess et al. 2000; BHAC15) via `IsocFit` (Baraffe et al., 2015). 3. Train, optimize (via Grid Search and K -fold cross-validation), and statistically evaluate machine learning regression algorithms (e.g., Random Forests, Support Vector Regressors, k -NN) to construct semi-empirical mass-magnitude relations. 4. Obtain stellar parameters including mass, age, effective temperature and luminosity.

State of the field

Several computational tools exist within the astronomical ecosystem for stellar parameter estimation and cluster analysis: * Machine learning tools applied to stellar parameter derivation, such as CARMENES transfer-learning routines (Bello-García et al., 2023), target specific spectral types (M dwarfs) rather than open clusters. * Empirical relations based on low-mass field binaries (Benedict et al., 2016; Mann et al., 2019) provide benchmarks for main-sequence objects, but lack integrated GUI workflows for cluster-wide missing-data imputation and pre-main-sequence fits.

SPECTRA fills this gap by combining GUI-driven visual diagnostics (e.g., Hertzsprung-Russell and diagnostic residual plots), automated missing-data imputation, and a model-ranking framework based on goodness-of-fit metrics ($RMSE$, MAE , R^2 , and AIC).

Software design and workflow

SPECTRA is written in Python 3 and relies on core packages including `numpy`, `pandas`, `scikit-learn`, `statsmodels`, and `ttkbootstrap` for user interface rendering (Gomes, 2024). The application's architecture decouples GUI control logic from heavy computational backends, providing a modular design structured around three main analytical pathways.

Figure 1 illustrates the high-level execution flow and operational logic of SPECTRA:

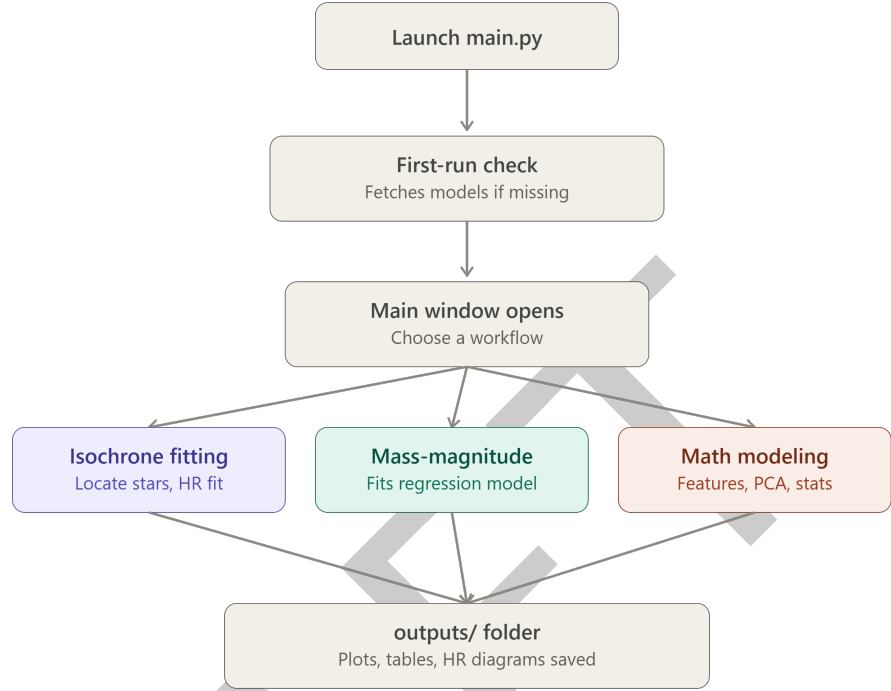


Figure 1: Execution workflow and modular architecture of SPECTRA.

Execution Pipeline & Module Description

1. **Initialization & Environment Setup:** Upon launching `main.py`, SPECTRA performs an automated first-run verification check (Gomes, 2024). If local theoretical evolutionary grids (e.g., Siess et al. 2000; BHAC15) or model weights are missing, the system automatically fetches and caches them to ensure reproducible offline execution (Baraffe et al., 2015; Gomes, 2024).

2. **Core Analytical Workflows:** From the primary desktop window, users can direct multi-catalog photometry (e.g., *Gaia* DR3, TESS) into three specialized processing modules (Babusiaux et al., 2023; Gomes, 2024; Paegert et al., 2021; Stassun et al., 2019):

- **Isochrone Fitting Module (IsocFit):** Implements local grid-search minimization to map observed stars onto theoretical evolutionary tracks (Gomes, 2024). The Euclidean distance δ in the HR diagram space is minimized via:

$$\delta = \sqrt{(T_{\text{eff}} - T_{\text{eff},0})^2 + (L - L_0)^2}$$

where $(T_{\text{eff},0}, L_0)$ are interpolated target parameters, followed by refined local track interpolation across adjacent stellar mass and age boundaries (Gomes, 2024).

- **Mass-Magnitude Modeling Module (RMM):** Builds semi-empirical relations by training candidate regression algorithms (e.g., k -NN, Random Forests, Support Vector Regressors, Bayesian Linear Regression) over theoretical stellar grids (Gomes, 2024). Hyperparameters are optimized via K -fold cross-validation and Grid Search (Gomes, 2024). Candidate models are evaluated and automatically ranked using a normalized multi-metric score S :

$$S = \hat{\mathcal{M}} \cdot \mathcal{P}^T$$

where $\hat{\mathcal{M}}$ represents the min-max normalized metrics matrix and $\mathcal{P}$ corresponds to an unit vector, assigning equal relative importance to $RMSE$, MAE , $(1 - R^2)$, and AIC to compute the composite metric score via matrix multiplication (Gomes, 2024).

- **Mathematical Modeling Module:** Provides advanced preprocessing, dataset filtering, missing attribute imputation (e.g., k -NN and iterative stochastic techniques), feature correlation analysis, Principal Component Analysis (PCA), and residual diagnostic checks (homoscedasticity and normality of errors) (Gomes, 2024).

3. **Output Artifact Generation:** All generated diagnostics—including annotated HR/Color-Magnitude diagrams, statistical summary tables (.csv), regression report logs, and diagnostic residual plots—are automatically organized and saved in the localized outputs/repository for direct export into publications (Gomes, 2024).

Research impact & empirical case study

The practical effectiveness of SPECTRA was validated through a comprehensive study of young stellar clusters covering distinct evolutionary stages: Orion Nebula Cluster (ONC, ~ 1 Myr) (Hillenbrand, 1997), Upper Scorpius (~ 5 – 11 Myr) (Pécaut et al., 2012; Rebull et al., 2018), h Persei / NGC 869 (~ 13 Myr), Pleiades (~ 112 Myr) (Dahm, 2015; Gossage et al., 2018; Lodieu et al., 2019), and NGC 2516 (~ 150 Myr).

Methodological Validation

Using a control sample of isolated eclipsing binary stars from Benedict et al. (2016), SPECTRA's k -NN regression model derived from the BHAC15 isochrone grid (Baraffe et al., 2015) achieved high predictive accuracy ($R^2 = 0.96$, $P_{\text{skedasticity}} = 0.12$) with a median residual uncertainty of 3.8%.

Cluster Analysis Results

- **Young PMS Clusters (ONC & Upper Scorpius):** For clusters rich in pre-main-sequence objects with significant age dispersion (Hillenbrand, 1997; Pécaut et al., 2012), IsocFit successfully identified low-mass PMS objects, whereas k -NN regression provided robust mass estimates with less dependency on bolometric corrections (Pécaut & Mamajek, 2013).
- **Well-Defined Main-Sequence Clusters (Pleiades & NGC 2516):** On clusters with established main sequences (Lodieu et al., 2019), k -NN RMM modeling using *Gaia* G -band and V -band absolute magnitudes yielded near-perfect alignment with literature values ($R^2 > 0.99$, $RMSE < 0.01 M_{\odot}$), demonstrating superior performance over traditional isochrone grid edge-effects.

Figure 2 illustrates a comparative evaluation of mass estimates derived by SPECTRA for member stars of the Pleiades cluster using both the traditional grid-interpolation method (IsocFit - BHAC15, left panels) and the machine-learning Mass-Magnitude relation (KNN - BHAC15, right panels), benchmarked against literature values from TESS V8.2 (Stassun et al., 2019), Lodieu et al. (2019) (Lodieu et al., 2019), and Delfosse et al. (2000) (Benedict et al., 2016).

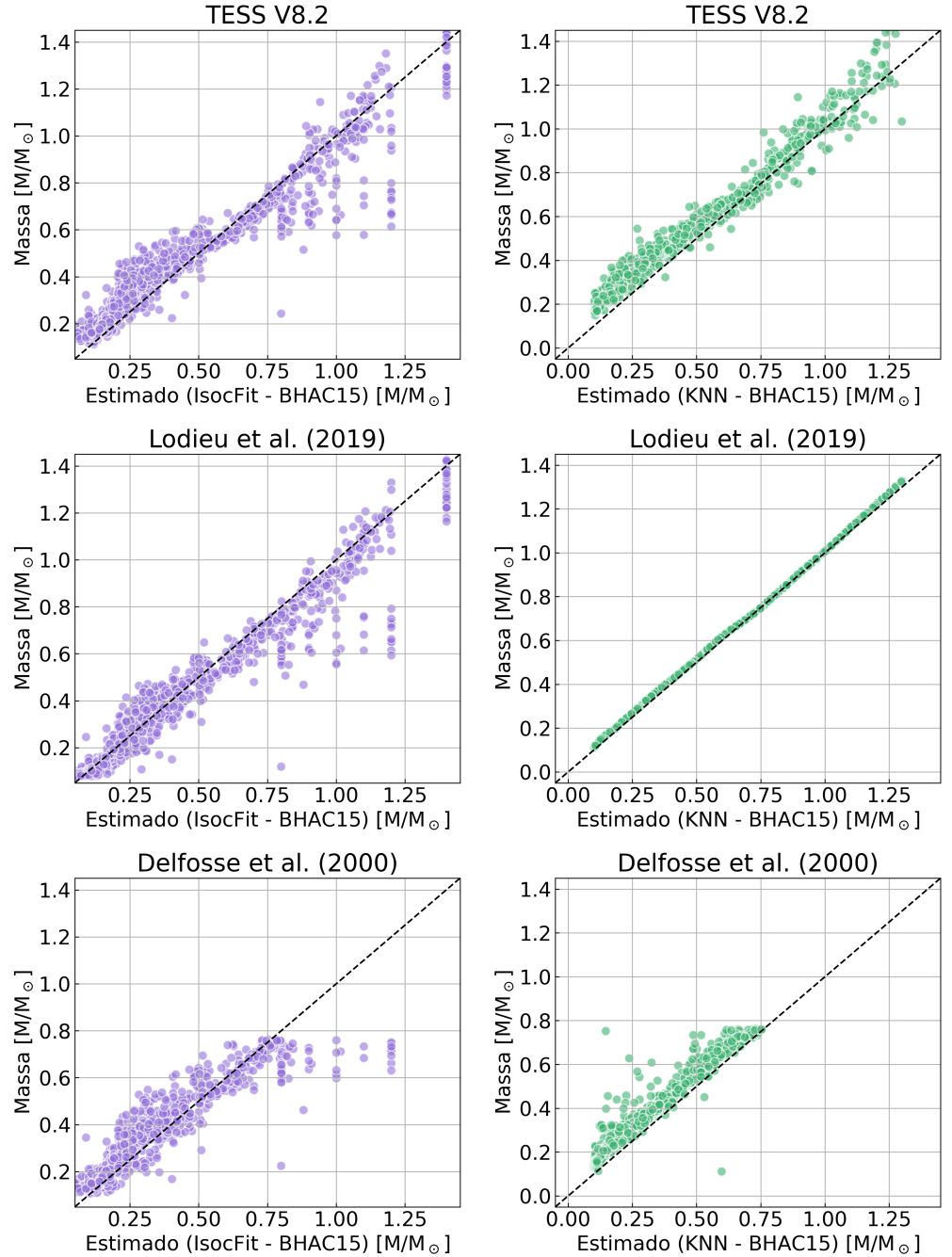


Figure 2: Comparison of stellar mass estimates for the Pleiades cluster obtained via IsocFit (left column) and KNN - BHAC15 (right column) against reference datasets from TESS V8.2, Lodieu et al. (2019), and Delfosse et al. (2000). The dashed line represents the ideal 1:1 identity relation.

As shown in Figure 2, the IsocFit - BHAC15 method exhibits greater dispersion around the 1:1 identity line, particularly for stars with $M > 0.7M_{\odot}$. This behavior is attributed to the convergence of evolutionary tracks near the Zero-Age Main Sequence (ZAMS) at the age of the Pleiades (~ 112 Myr), which creates localized degeneracies during grid minimization [cite: 1]. Conversely, the semi-empirical KNN - BHAC15 model demonstrates exceptional predictive stability, showing near-perfect alignment ($R^2 \approx 1.0$) when cross-validated against the photometrically derived masses of Lodieu et al. (2019). When benchmarked against empirical

binary masses from Benedict et al. (2016), the k -NN model reproduces stellar masses with minimal bias up to the applicability threshold of the empirical K -band relation ($M \lesssim 0.75M_{\odot}$) [cite: 1].

A more detailed astrophysical discussion and complete parameter catalogues derived for each cluster will be presented in a forthcoming paper (Gomes et al., in prep.).

By reducing manual preprocessing time and providing rigorous statistical diagnostic reports, SPECTRA supports both observational research in stellar rotation/dynamics and undergraduate/graduate instruction in stellar astrophysics (Gomes, 2024).

AI usage disclosure

No generative AI tools were used in the algorithmic development or code implementation of the SPECTRA package.

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